

Detailed Software Requirements Specification (SRS)

Sales KPI Performance & Invoice Approval Management System

1. Project Vision

Build a cloud-based KPI Performance Management System for sales organizations. The platform measures employee performance using assigned KPIs and approved sales invoices. The system must provide real-time visibility into individual, team, branch, supervisor, and company performance.

2. User Roles

System Administrator: Full control. Manager: Assign KPIs, view company performance, leaderboards, reports. Supervisor: Review and approve sales invoices. Sales Personnel: Submit invoices and monitor personal KPIs. Executive Viewer: Read-only access to executive dashboards.

3. KPI Management

Create KPI templates, KPI categories, weighted KPIs, monthly targets, branch targets, team targets, and individual targets. Managers can assign KPIs individually or in bulk.

4. Invoice-Based Achievement Tracking

Sales personnel create sales records using invoice submission. Fields include invoice number, invoice date, customer, amount, description, image attachment, and optional notes. Draft and submit workflow required.

5. Approval Workflow

Submitted invoices are routed to supervisors. Actions: Approve, Reject, Request Clarification. Only approved invoices contribute to KPI achievement. Every action must be logged in an audit trail.

6. KPI Calculation Engine

Achieved KPI = Sum of approved invoices. Performance Percentage = $\text{Achieved KPI} / \text{Assigned KPI} \times 100$. Support weighted KPI calculations and overall employee score.

7. Personnel Dashboard

Show assigned KPIs, approved sales, pending sales, rejected sales, achievement percentage, performance status, monthly trends, ranking, forecasted month-end performance, and required daily sales.

8. Supervisor Dashboard

Pending approvals queue, approval statistics, invoice review screen, employee performance summaries, approval history, and team performance.

9. Manager Dashboard

Company target vs achieved. Branch comparison. Supervisor comparison. Top performers. Bottom performers. Pending approvals. Monthly trends. Performance forecasting. Drill-down into employee

profiles.

10. Executive Dashboard

High-level company performance. Total target. Total achieved. Achievement percentage. Branch ranking. Supervisor ranking. Top 10 sales personnel. Bottom 10 requiring attention. Forecasted end-of-month results.

11. Leaderboard Module

Top sales personnel by revenue. Top KPI achievers. Most approved invoices. Most improved employee. Top branches. Top supervisors. Personal ranking visible to every salesperson.

12. TV / Display Mode

Auto-rotating presentation screen. Displays one employee at a time. Automatically changes every 2 minutes. Shows photo, achievement, target, status, ranking, recent sales, forecast, and countdown timer. Supports fullscreen mode for office displays.

13. Employee Profile Card

Profile photo. Placeholder avatar when photo is unavailable. Name. Role. Branch. Supervisor. Rank. Current achievement. Performance trend.

14. Countdown Timer

Live countdown showing days, hours, minutes, and seconds remaining until month-end. Visible on manager, executive, TV display, and salesperson dashboards.

15. Performance Health Indicators

Green = On Track. Amber = At Risk. Red = Behind Schedule. Calculated from KPI achievement versus elapsed time in the month.

16. AI Forecasting

Predict month-end achievement. Estimate target completion likelihood. Calculate required daily sales. Identify employees at risk of missing targets.

17. Notifications

Invoice approval notifications. Rejected invoice alerts. Target achievement alerts. Leaderboard position changes. Month-end reminders.

18. Reports

Employee performance report. Supervisor report. Branch report. Company report. Leaderboard report. Invoice approval report. Export PDF, Excel, and CSV.

19. Audit Trail

Track invoice submissions, approvals, KPI assignments, profile changes, login activity, and report generation.

20. Database Entities

Users, Roles, Departments, Branches, KPI Definitions, KPI Assignments, Invoices, Invoice Attachments, Approvals, KPI Logs, Notifications, Leaderboards, Reports, Audit Logs.

21. Security Requirements

JWT authentication. Role-based access control. Password encryption. Activity logging. Secure file uploads. Data backup and recovery.

22. Technical Architecture

Frontend: React/Next.js. Backend: NestJS. Database: PostgreSQL. Storage: Cloud object storage. Authentication: JWT. Deployment: Docker and cloud infrastructure.

23. Important Features (Must Have)

1. KPI Assignment. 2. Bulk KPI Assignment. 3. Invoice Submission. 4. Invoice Image Upload. 5. Supervisor Approval Workflow. 6. Automatic KPI Calculation. 7. Employee Dashboard. 8. Manager Dashboard. 9. Executive Dashboard. 10. Leaderboard. 11. TV Display Mode. 12. Employee Photos. 13. Countdown Timer. 14. Performance Health Indicators. 15. AI Forecasting. 16. Reports & Exports. 17. Notifications. 18. Audit Trail. 19. Branch Performance Tracking. 20. Supervisor Performance Tracking.

24. AI Agent Deliverables

Generate database schema, API documentation, frontend screens, responsive dashboards, role permissions, reporting engine, invoice workflow, leaderboard engine, TV display module, forecasting services, automated tests, and deployment scripts.